COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
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NAME:	Libo-on, Gian Carlo G.	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. **Prelim Examination**

1. PROCEDURE OUTPUT

```
//matching pairs
const std::unordered_map<char, char> pair = {
    {'1', '1'},
    {'2', '2'},
    {'3', '3'},
    {'4', '4'},
    {'5', '5'},
    {'6', '6'},
    {'7', '7'},
    {'8', '8'},
    {'9', '9'},
    {'0', '0'},
};

//opening characters
bool isOpening (char ch) {
    return ch == '1' || ch == '2' || ch == '3' || ch == '4' || ch == '5' || ch == '6' || ch == '7' || ch == '8' || ch == '9' || ch == '0';
}
//closing characters
bool isClosing (char ch) {
    return pair.count(ch) > 0;
}
```

Code Snippet for Delimiters and Matching Pairs

```
//array checking if the opening and closing characters match
bool checkArray(const std::string& expr){
    Array stack;
    for (char ch : expr) {
        if (isOpening(ch)){
            stack.push(ch);
        } else if (isClosing(ch)){
            if (stack.isEmpty() || stack.pop() != pair.at(ch)) {
                return false;
            }
        }
    }
    return stack.isEmpty();
}
```

Code Snippet for Checking if Opening and Closing Characters Match Using Array

```
/linked list checking if opening and closing matches
   bool checkLinkedList(const std::string& expr){
       LinkedList stack;
       for (char ch : expr){
           if (isOpening(ch)){
               stack.push(ch);
           } else if (isClosing(ch)) {
               if (stack.isEmpty() || stack.pop() != pair.at(ch)){
                   return false;
               }
           }
       }
       return stack.isEmpty();
   }
```

Code Snippet for Checking if Opening and Closing Characters Match Using Linked List

```cpp
int main() {

//test cases
    std::string testCases[] = {
        "12321",
        "12345",
        "1221",
        "7",
        "123321",
        "1234325"
    };

    std::cout <<  "====================Symmetrical Testing====================\n\n";

    std::cout << std::left
              << std::setw(22) << "Sample"
              << std::setw(16) << "Array Stack"
              << std::setw(22) << "Linked List Stack"
              << "\n";
    std::cout <<  "===========================================================\n";
//displaying results
    for (const std::string& expr : testCases) {
        bool resArr = checkArray(expr);
        bool resLL =  checkLinkedList(expr);

        std::cout << std::left
                  << std::setw(22) << expr
                  << std::setw(16) << (resArr ? "Symmetrical" : "Asymmetrical")
                  << std::setw(22) << (resLL ? "Symmetrical" : "Asymmetrical")
                  << "\n";
        }
    return 0;
}
```

Code Snippet for Displaying results

```
====================Symmetrical Testing====================

Sample                Array Stack     Linked List Stack
===========================================================
12321                 Asymmetrical    Asymmetrical
12345                 Asymmetrical    Asymmetrical
1221                  Asymmetrical    Asymmetrical
7                     Asymmetrical    Asymmetrical
123321                Asymmetrical    Asymmetrical
1234325               Asymmetrical    Asymmetrical
```

Output for Failed Output

Input Sequence	Symmetric?	Output (Console)	Remarks
1 2 3 2 1	Yes	Asymmetrical	Failed
1 2 3 4 5	No	Asymmetrical	Failed
1 2 2 1	Yes	Asymmetrical	Failed
7	Yes	Asymmetrical	Failed
1 2 3 3 2 1	Yes	Asymmetrical	Failed
1 2 3 4 3 2 5	No	Asymmetrical	Failed

Pseudo code

Array Stack
Define max size
Construct Matching pairs and delimiters (use characters)
Create isOpening function (should be able to read the first character of the string)
Create isClosing function (should be able to read the last character of the string)
Create the Array class
Indicate the top index as size -1
Create the push function (checks if the stack is full, if not it should increment a unit to the current value)
Create the pop function (checks if the stack is empty, if not it should decrement a unit to the current value)
Create the top function (checks the character at the top of the stack)
Create the isEmpty function (should be able to check if the stack is empty if array size is equal to -1)
Create the checking function (should be able to check if the opening and closing character matches)

Linked List Stack
Construct the Node (should be able to detect if node is a character and is able to point to the next node)
Create the Linked List class
Create the push function (adds a new node and points to the new node, new node should be the top node)
Create the pop function (checks if the list is empty and returns a null value, Creates a temporary node that becomes the top node and pushes the data to the next node, after that, the temporary node is deleted.)
Create the isEmpty function (should be able to check if the stack is empty if top node is equal to null)
Create the checking function (should be able to check if the opening and closing character matches)

Array Queue
Create a header file for Array Queue implementation
Define array max size
Create a Class for the Array (use characters)
Create a front function (Checks the first index)
Create a back function (checks the back index)
Create a check function (Checks if the data in the indexes are symmetrical to each other)

Create a traversal function (travels from the front index to the back index)
Create an enqueue function (inserts a data in the back of the array, follows the FIFO rule)
Create a dequeue function (only removes the data in the first index of the array)
Create an isEmpty Function (checks if the array is empty)
Create a display function (displays the results of the tests)
Create a main file (imports the header files and its functions)

Linked List Queue
Create a header file for the Linked List Queue implementation
Create a Class Node (use characters for easier comparison)
Create a node that points to a new node (when data is added then point the new node next to the nullptr
Create an enqueue function (it should also be able to follow the FIFO rule and inserts only at the back of the list.)
Create a dequeue function (if a node has to be deleted, a temporary node should be created as a place holder and is removed when the operation is done. The next node will then become the new top node)
Create an isEmpty function (only for checking if the queue is empty)
Create a checking function (which checks if the data in the nodes are symmetrical)
Create a traversal function (which checks each node that is in the queue)
Create a display function (to display the value in each nodes and verifies if the data is symmetrical or not)
Create a main cpp file (allows you to import and call the functions listed in the linked list header file)

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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